THE LIMIT IS RECOVERED FROM A GOOD DIAGRAM

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5304-5321, following the statement at 5286-5302.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Scheme theoretic image commutes with etale localization. Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-scheme-theoretic-image (cross-chapter, cited at ~5307-5309). Reduces to W affine mapping into an affine U_i etale over B_i.
- Scheme-level good diagram limit. Limits, Lemma limits-lemma-limit-from-good-diagram (cross-chapter, cited at ~5319-5321). Concluding engine for the affine scheme case.

Source reference. Original equation 0CP8: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5264–5264. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CP9: focused-seeds/geometry/S9-spaces-limits-limit-from-good-diagram/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CP6: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5249–5249. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 082Z: focused-seeds/geometry/S9-spaces-limits-limit-from-good-diagram/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 0CP6. Let $X \to B$ be a quasi-separated and finite type morphism of algebraic spaces. Given $i \in I$ and a diagram*

$$\begin{array}{ccc} X & \longrightarrow & W \\ \downarrow & & \downarrow \\ B & \longrightarrow & B_i \end{array}$$

as in (0CP8) for $i' \geq i$ let $X_{i'}$ be the scheme theoretic image of $X \to B_{i'} \times_{B_i} W$. Then $X = \lim_{i' \geq i} X_{i'}$.

Proof. Since X is quasi-compact and quasi-separated formation of the scheme theoretic image of $X \to B_{i'} \times_{B_i} W$ commutes with étale localization (Morphisms of Spaces, Lemma 082Z). Hence we may and do assume W is affine and maps into an affine U_i étale over B_i. Then

$$B_{i'} \times_{B_i} W = B_{i'} \times_{B_i} U_i \times_{U_i} W = U_{i'} \times_{U_i} W$$

where $U_{i'} = B_{i'} \times_{B_i} U_i$ is affine as the transition morphisms are affine. Thus the lemma follows from the case of schemes which is Limits, Lemma 0CP9. □